

IMX Lineage model

Geonovum Information model

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Abstract

This document specifies the IMX Lineage model. The IMX lineage model provenance information generated during an IMX orchestration process, building upon [\[IMX-Metadata\]](#) and [\[IMX-ModelMapping\]](#).

Status of this document

This is a working draft that can be changed, removed or replaced by other documents at any time. It is not a stable document.

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Versiebeheer

Dit document is aan verandering onderhevig. Het versiebeheer van het document geeft inzicht in wijzigen en de actualiteit ervan.

Versie	Datum	Status	Bewerking	Toelichting
0.1	dd-mm-yyyy	concept	Initieel document	"Kapstok" document aangemaakt

§ 1. Introduction

Kadaster and Geonovum are developing a mechanism to facilitate the semantic integration of decentralized data registers. A central API operates on an integrated semantic layer. The semantic layer constitutes of a coherent information model integrating several domain specific domain models. An API orchestration layer directs data communication to and from existing data registries.

Data retrieval is according to the information model of the semantic layer. Data retrieval will optionally include metadata about the lineage (provenance) of the data.

This document specifies the lineage model used and the mechanism introduced in [[IMX-Metadata](#)] to relate the lineage data as metadata to published data.

§ 1.1 Scope

The scope of this lineage model is:

- lineage related to data being published as a result of data integration. The lineage will describe source and process/operation data
- lineage data at the instance level of properties

§ 1.2 Working proces

This document, lineage model, is developed in the WaU project The principal working method is agile development through several High5 sessions on use cases of increasing complex information content and

§ 2. Analysis

§ 2.1 Context

In today's complex and interconnected world, the answers that people are seeking often require access to data from multiple registries. This is because the information needed to answer a question or solve a problem is often distributed across multiple sources and domains. Data needs to be aggregated and consolidated in order to satisfy user needs. This needs to be done transparently. It should be clear where the resulting data came from and what has been done to it. For this, insight into lineage of data should be provided.

The lineage model explained in this document provides a standardised way to work with, store and communicate lineage information of data.

§ 2.2 Related or referenced standards

- [\[IMX-ModelMapping\]](#)
- [\[prov-o\]](#)
- [\[IMX-Metadata\]](#)

§ 2.3 Relevant documentation

[\[Property-Stereotype-for-Metadata\]](#), Interactive Instruments

§ 3. Requirements and approach

Within this chapter two sections are presented; Requirements and Approach. In the Requirements section the requirements for the information model are listed. In the Approach section the requirements are translated into a conceptual model which forms the basis for the logical model described in [chapter 4](#).

§ 3.1 Requirements

The following requirements have been taken into account in the approach:

1. **We should be able to identify and describe singular data items**
2. **We should be able to describe which data items contributed to the generation of a data item**
3. **We should be able to describe activities that act upon data items**
4. **We should be able to find data items that contributed to the generation of a data item**
5. **We should be able to describe how data item contributed to the generation of a data item**

§ 3.2 Approach

The first requirement, to identify and describe singular pieces data, is realized by the introduction of the data item objecttype as described in [IMX-Metadata](#).

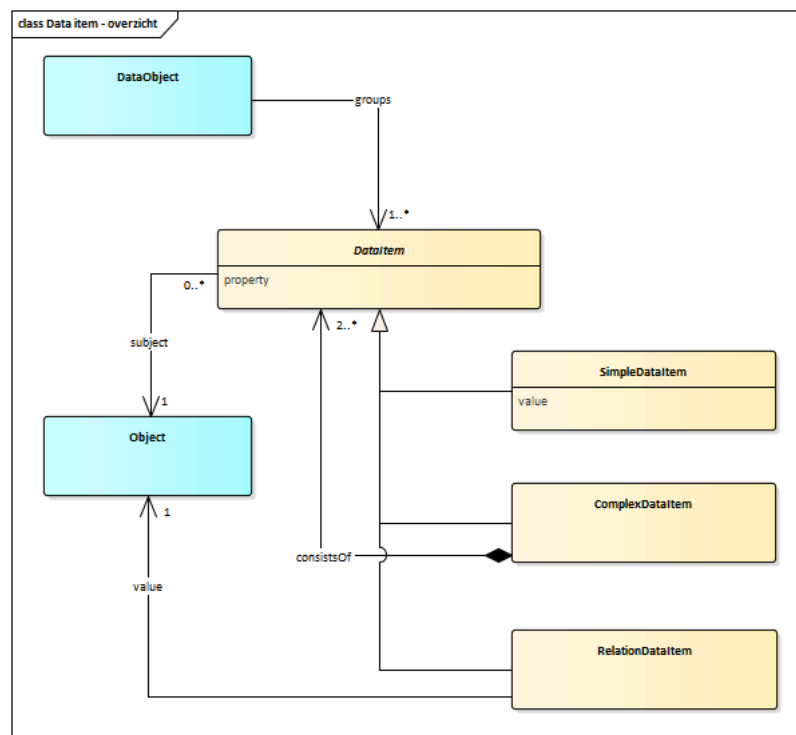


Figure 1 Data item

A [data item](#) can be described from different perspectives. This gives a basic model, applicable for many usecases. The perspective the lineage model takes is one of provenance [[prov-o](#)]. The lineage model adds to this a way to connect one data item to one or more other data items it [is derived from](#). The latter are called [source data items](#), the former [orchestrated data items](#). For the [source data item](#) it is important to know from which [data source](#) it is retrieved so that users can interpret the [source data item](#) in the context it was published in. This also allows allows users to find related information.

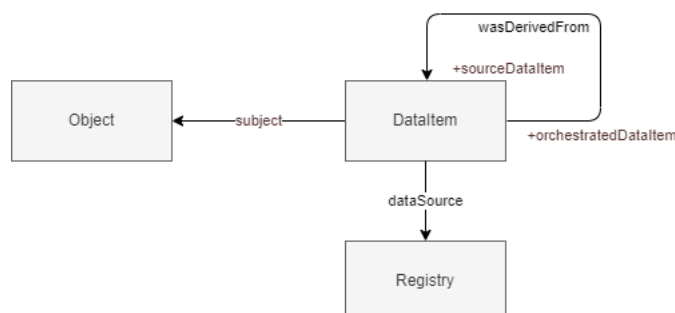


Figure 2 Orchestrated data items derived from source data items

Term	Definition
source data item	A source data item is a data item that serves as input for an orchestration activity .
orchestrated data item	An orchestrated data item is a data item that is generated/obtained by orchestrating source data.
data object	A data object is a contextually grouped set of data items .
data source	A data source is an entity which provides access to a dataset.

A data example:

```

{
  "identification": "G0200.42b3d39246840268e0530a0a28492340",
  "constructionDate": "2006",
  "registeredWith": {
    "consistOf": [
      {
        "property": "bouwjaar",
        "value": {
          "stringValue": "2006"
        }
      },
      "subject": {
        "identificatie": "G0200.42b3d39246840268e0530a0a28492340",
        "domein": null
      },
      "wasDerivedFrom": {
        "property": "oorspronkelijkBouwjaar",
        "value": {
          "stringValue": "2006"
        },
        "subject": {
          "identificatie": "0200100000085932",
          "type": "Pand"
        }
      }
    ]
  }
}

```

This, however, does not provide any insight into how the [source data item\(s\)](#) were used to generate the [orchestrated data item](#). In order to allow for this the generation itself is identified as an [orchestration activity](#), which is a type of [activity](#). This provides the information that the [data item](#) is generated using orchestration techniques.

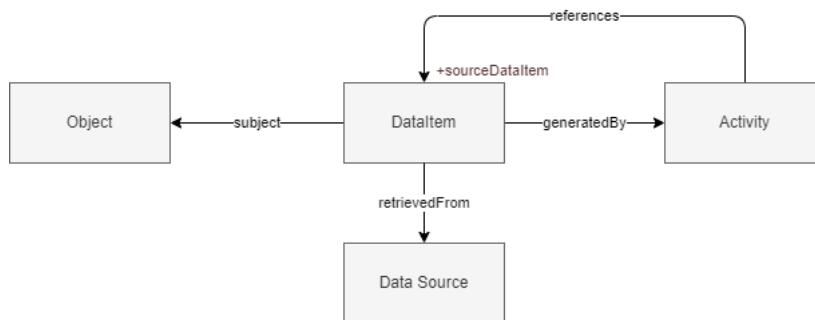


Figure 3 A data item is generated by an activity


```

{
  "identification": "G0200.42b3d39246840268e0530a0a28492340",
  "constructionDate": "2006",
  "registeredWith": {
    "consistsOf": [
      {
        "property": "bouwjaar",
        "wasGeneratedBy": {
          "used": {
            "property": "oorspronkelijkBouwjaar",
            "value": {
              "stringValue": "2006"
            },
            "subject": {
              "identificatie": "0200100000085932",
              "type": "Pand"
            }
          }
        },
        "value": {
          "stringValue": "2006"
        },
        "subject": {
          "identificatie": "G0200.42b3d39246840268e0530a0a28492340",
          "domein": null
        }
      }
    ]
  }
}

```

Lastly, we provide insights into the transformation itself, the mapping used for the orchestration activity. The mapping is described in a separate artifact called the [model mapping](#). This document describes the relation between two models and how data from one can be transformed to the other. The mapping is structured according to the [mapping specification](#). The rules to generate a specific [data item](#) are encompassed in a [property mapping](#), which is executed during orchestration. To describe this, we introduce a specific type of [orchestration activity](#), the [property mapping execution](#).

Term	Definition
<i>used</i>	Usage is the beginning of utilizing an entity by an activity . Before usage, the activity had not begun to utilize this entity and could not have been affected by the entity. (based on prov:used)
<i>orchestration activity</i>	An orchestration activity is an activity that is part of an IMX orchestration process.
<i>property mapping execution</i>	A Property mapping execution is an orchestration activity that pertains to the execution of a property mapping .

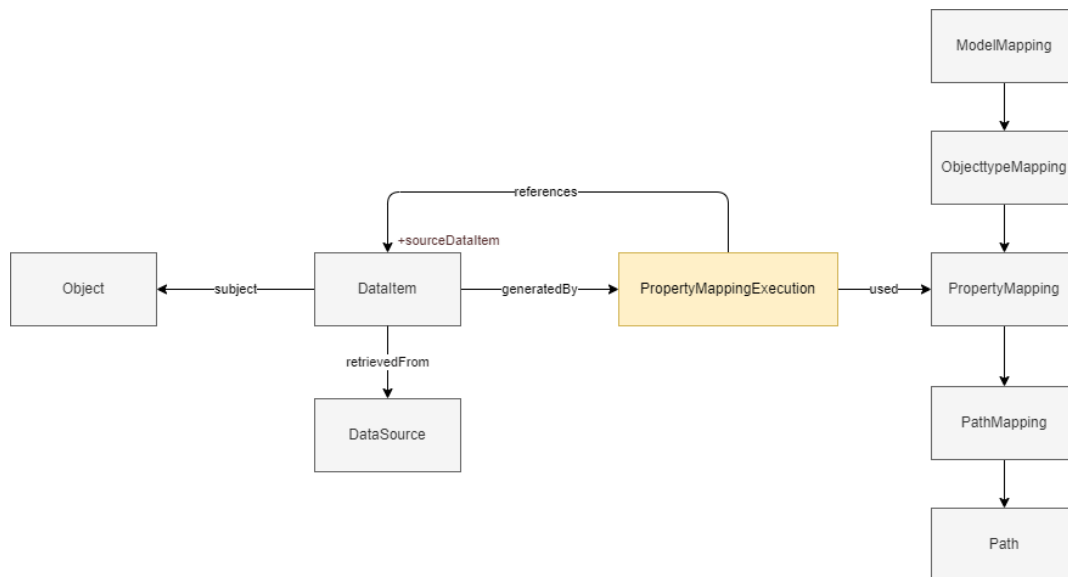


Figure 4 A data item is generated by a property mapping execution

The [property mapping execution](#) can be deconstructed into several smaller activities as it may include multiple intermediary results; each having a query [path](#) and a [start node](#), which are then combined and/or transformed into the [value](#) of the [orchestrated data item](#). The [PropertyMapping](#) is the main entity describing the steps ([path mappings](#)) required to derive a specific data item in order of execution. Each step is an activity that informs the [property mapping execution](#); which yields the [orchestrated data item](#). As part of the lineage information we refer to elements in the mapping to provide insights into how the [orchestrated data item](#) was constructed. This allows us to convey which data items were used to derive a new data item, and how these were used. Adding this to our lineage model yields the following model.

Term	Definition
was informed by	A relation between activities stating that one activity uses some output of the other activity. (based on prov:wasInformedBy)
path mapping execution	A property path execution is an activity that retrieves data items to serve as input for the property mapping execution .
references	The source data item referenced by following the defined path starting from the start node .
start node	The source object from which the path is followed to find the used data item.

A [path mapping](#) consists out of one [path](#); possibly yielding any number of [data items](#), when executed. The lineage should give insights on precisely how each [data item](#) was found and how it contributed to the generation of the [orchestrated data item](#). For each "OrchestrationActivity -used-> DataItem" we map to the respective [path](#).

We do this by further deconstructing the [path mapping execution](#) into each [path](#) that is defined in the mapping. For each [path execution](#) that informs the [path mapping execution](#) a [used](#) property is linked to the respective [path](#) in the [model mapping](#). This allows us to point to the exact mapping element providing the path that leads us to the [source data item](#). A pointer to the node from which the path is followed is added, this is the [start node](#).

Term	Definition
Path execution	A path execution is an orchestration activity that is part of the path mapping execution .
source object	A source object is an object that that is referenced to create an orchestrated data item .

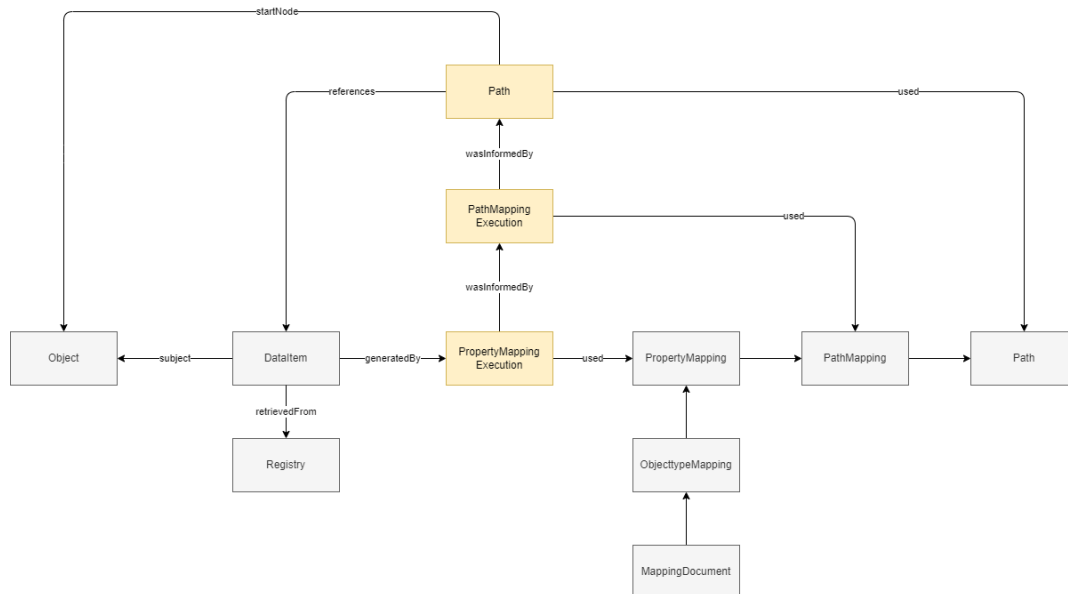


Figure 5 A data item is generated by a property mapping execution, referencing other data items

In this model several types have different roles following from incoming relations. This could make the datamodel less clear. In the following diagram roles are made explicit.

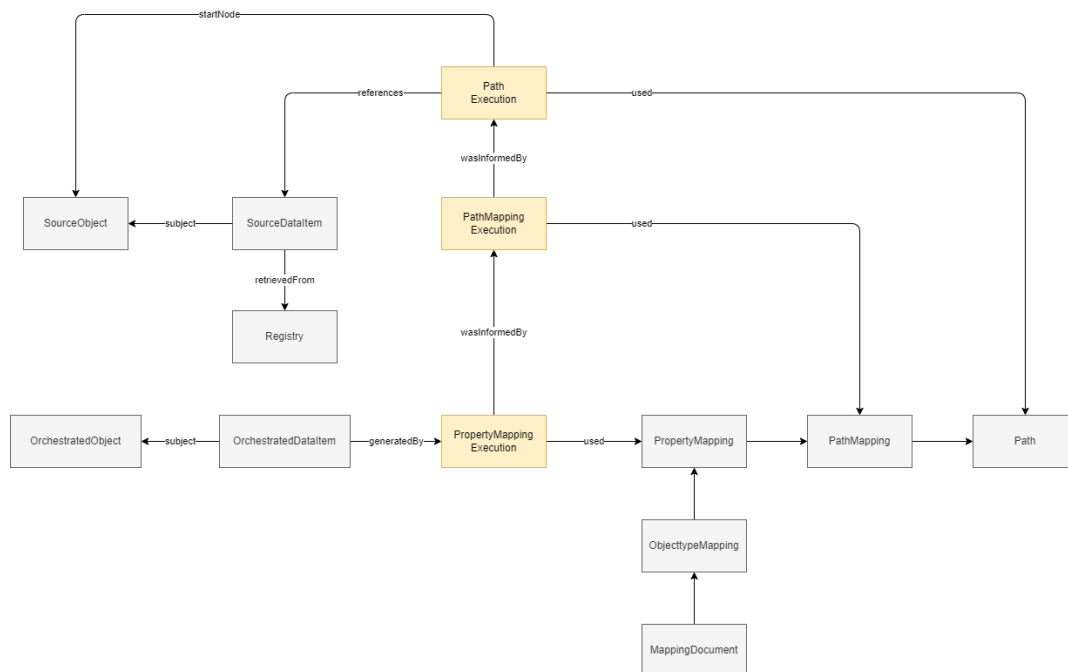


Figure 6 Orchestrated data item is generated by a property mapping execution, referencing source data items

In JSON this could look as follows.

```

{
  "identification": "G0200.42b3d39246840268e0530a0a28492340",
  "constructionDate": "2006",
  "hasLineage": {
    "orchestratedProperties": [
      { <!-- OrchestratedDataItem -->
        "property": "bouwjaar",
        "value": {
          "stringValue": "2006"
        },
        "subject": {
          "identificatie": "G0200.42b3d39246840268e0530a0a28492340",
          "domein": null
        },
        "wasGeneratedBy": { <!-- PropertyMappingExecution -->
          "used" : { <!-- property mapping -->
            "objectKey": "MAP.IMXGEO.000102",
            "objectType": "PropertyMapping"
          }
          "wasInformedBy": { <!-- PathMappingExecution -->
            "references": { <!-- source data item -->
              "property": "oorspronkelijkBouwjaar",
              "value": {
                "stringValue": "2006"
              },
              "subject": {
                "objectKey": "0200100000085932",
                "objectType": "Pand"
              }
            },
            "used": { <!-- Path -->
              "path": {
                "segments": [
                  "isGerelateerdAan",
                  "oorspronkelijkBouwjaar"
                ]
              }
            },
            "startNode": { <!-- SourceObject -->
              "objectKey": "G0200.42b3d39246840268e0530a0a28492340",
              "objectType": "Pand"
            }
          }
        }
      }
    ]
  }
}

```

When multiple source data items are used this would result in multiple "wasInformedBy": { ... } statements within the "wasGeneratedBy":{ ... } element.

§ 3.2.1 Summary

The requirements are reflected in the model by introducing the following concepts:

- **Data item**

- "orchestrated data item" for req 1
- "source data item" for req 2.

In order to be able to talk about them.

- **Orchestration Activity**

- Property Mapping execution and propertyMapping execution for req 3

In order to describe that orchestration took place.

- **Data source** for req 4.

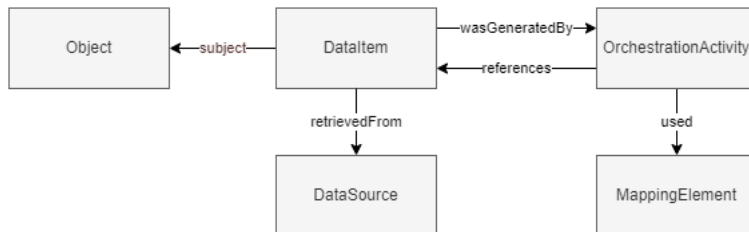
In order to describe where source data is retrieved from and thus can be found.

- **Mapping element**

- Property Mapping and path for req 5.

In order to precisely indicate what rules were applied to the source data to derive the orchestrated data item(s).

In short:

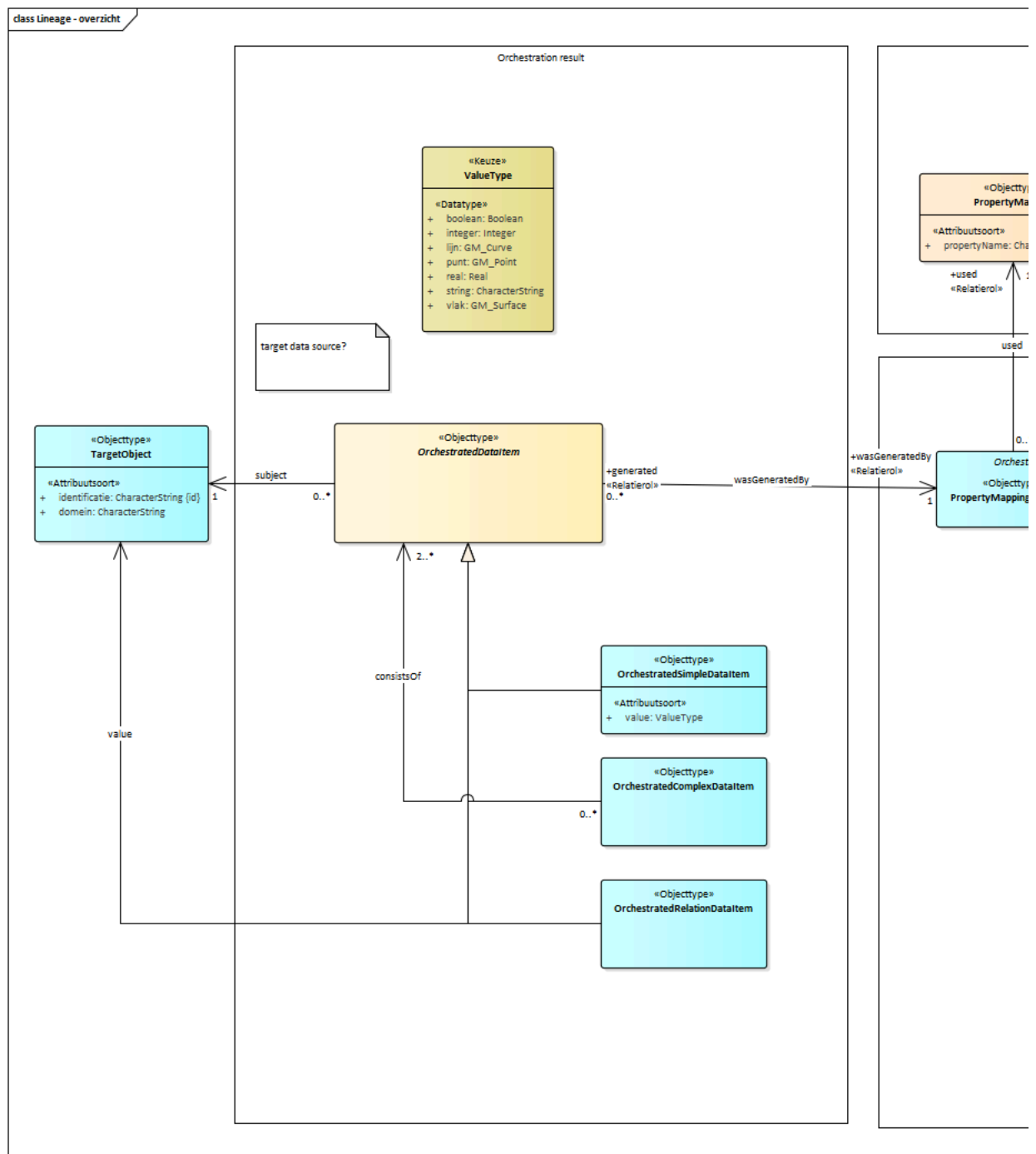


§ 3.2.2 Limitations

- We don't indicate the order of the activities or path segments which is an important aspect of the mapping
- We don't have a way of describing the (software) agent responsible for the orchestration (can be copied from PROV-O).
- We don't describe objecttype mapping executions.
- ...

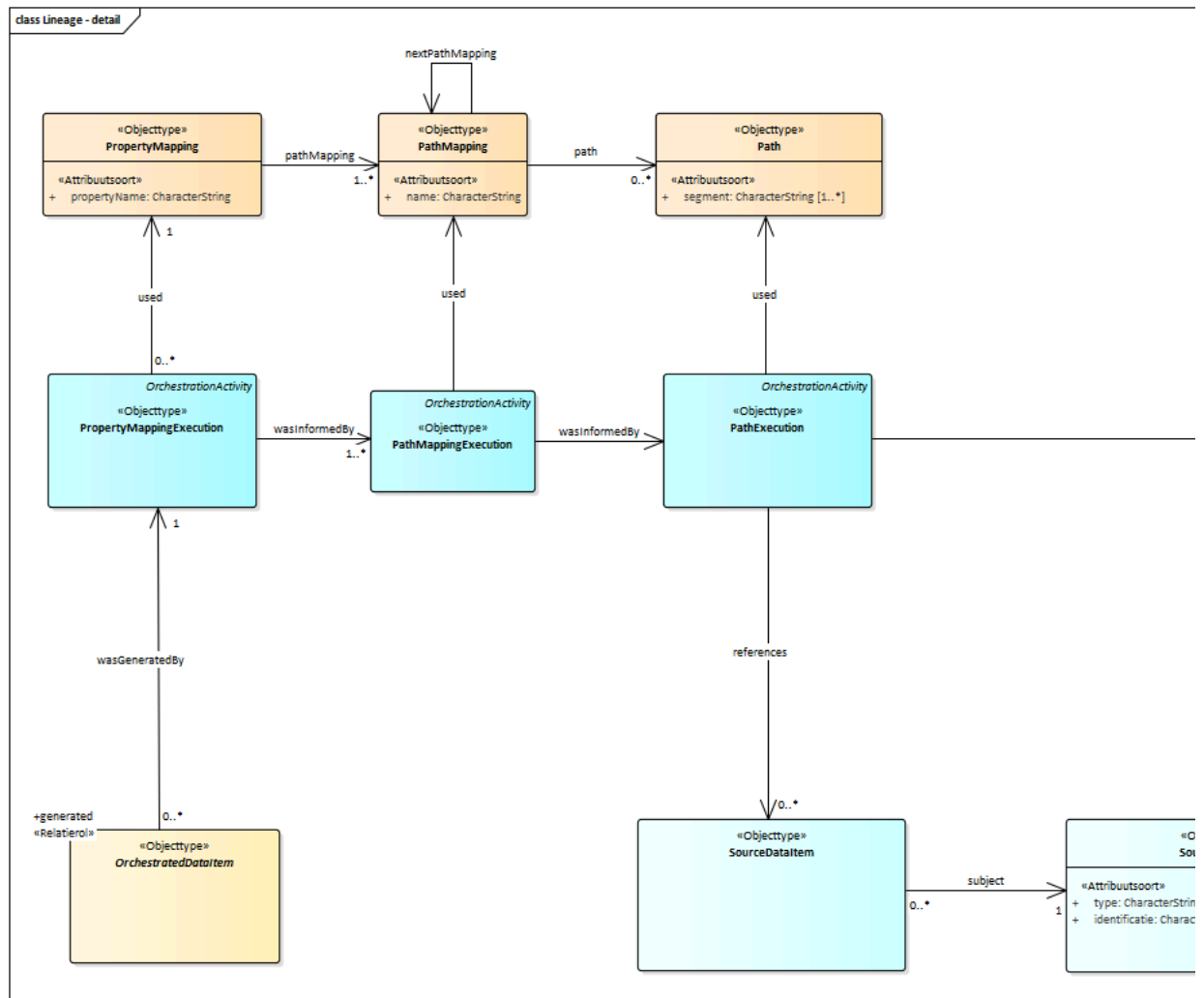
§ 4. Information model and Objectcatalogue

§ 4.1 Information model



Lineage

§ 4.2 Domain Orchestration



§ 4.2.1 Objecttypes

§ 4.2.1.1 OrchestratedDataItem

Name	OrchestratedDataItem
Definition	An orchestrated data item is a data item that is generated/obtained by orchestrating source data.
Is abstract	True

§ OVERVIEW RELATIONS

Relation name with multiplicity	Definition
OrchestratedDataItem [0 .. *] <u>wasGeneratedBy: was-GeneratedBy</u> PropertyMappingExecution [1]	The activity which generated (or produced) the orchestrated data item.

OrchestratedDataItem [0 .. *] subject TargetObject [1]	The subject of the data item.
OrchestratedComplexDataItem [0 .. *] consistsOf OrchestratedDataItem [2 .. *]	The constituent data item of the complex data item.

§ 4.2.1.2 *SourceDataItem*

Name	SourceDataItem
Definition	A source data item is a data item that serves as input for an orchestration activity.
Is abstract	True

§ OVERVIEW RELATIONS

Relation name with multiplicity	Definition
SourceDataItem [0 .. *] dataSource Data Source [1]	The data source from which the source data item can be retrieved
SourceDataItem [0 .. *] subject SourceObject [1]	The subject of a source data item.
PathExecution [1] references SourceDataItem [0 .. *]	The source data item referenced by following the defined path starting from the start node.
SourceComplexDataItem [1] consistsOf SourceDataItem [2 .. *]	The constituent data item of the complex data item.

§ 4.2.1.3 *SourceRelationDataItem*

Name	SourceRelationDataItem
Definition	A source relation data item is a relation data item that serves as input for an orchestration activity.
Is abstract	False

§ OVERVIEW RELATIONS

Relation name with multiplicity	Definition
SourceRelationDataItem [1] value SourceObject [1]	The value of the data item.
SourceRelationDataItem is specialisatie van SourceData-Item	A source data item is a data item that serves as input for an orchestration activity.

§ 4.2.1.4 SourceObject

Name	SourceObject
Definition	A source object is an object that serves as input for an orchestration activity.
Unique designation	identificatie
Is abstract	False

§ OVERVIEW ATTRIBUTES

Attribute name	Definition	Format	Card
<u>type</u>	The type of the source object in the data source.	<u>Character-</u> <u>String</u>	1
<u>identificatie</u>	The identification of the source object in the data source.	<u>Character-</u> <u>String</u>	1

§ OVERVIEW RELATIONS

Relation name with multiplicity	Definition
<u>SourceDataItem</u> [0 .. *] <u>subject</u> SourceObject [1]	The subject of a source data item.
<u>SourceRelationDataItem</u> [1] <u>value</u> SourceObject [1]	The value of the data item.
<u>PathExecution</u> [1] <u>startNode: startNode</u> SourceObject [1]	The source object from which the path is followed to find the referenced data item.

§ 4.2.1.5 Data Source

Name	Data Source
Definition	A data source is an entity which provides access to a dataset.
Is abstract	False

§ OVERVIEW ATTRIBUTES

Attribute name	Definition	Format	Card
<u>name</u>	The name of the data source.	<u>Character-</u> <u>String</u>	1

§ OVERVIEW RELATIONS

Relation name with multiplicity	Definition
Data Source [1] conformsTo Model [1]	
SourceDataItem [0 .. *] dataSource Data Source [1]	The data source from which the source data item can be retrieved

§ 4.2.1.6 OrchestratedRelationDataItem

Name	OrchestratedRelationDataItem
Definition	An orchestrated relation data item is an orchestrated data item that is non-divisible and where the value is an association to an object.
Is abstract	False

§ OVERVIEW RELATIONS

Relation name with multiplicity	Definition
OrchestratedRelationDataItem [1] value TargetObject [1]	
OrchestratedRelationDataItem is specialisatie van OrchestratedDataItem	An orchestrated data item is a data item that is generated/obtained by orchestrating source data.

§ 4.2.1.7 TargetObject

Name	TargetObject
Definition	A target object is an object which is the subject of an orchestrated data item.
Unique designation	identificatie
Is abstract	False

§ OVERVIEW ATTRIBUTES

Attribute name	Definition	Format	Card
identificatie	The identification of the target object	Character-String	1
Domain		Character-String	1

Relation name with multiplicity	Definition
OrchestratedDataItem [0 .. *] subject TargetObject [1]	The subject of the data item.
OrchestratedRelationDataItem [1] value TargetObject [1]	

§ 4.2.1.8 *PathExecution*

Name	PathExecution
Definition	A path execution is an orchestration activity that is part of the path mapping execution.
Date added	<date>
Is abstract	False

Relation name with multiplicity	Definition
PathExecution [1] references SourceDataItem [0 .. *]	The source data item referenced by following the defined path starting from the start node.
PathExecution [1] used: used Path [1]	The mapping element used in the orchestration activity.
PathExecution [1] startNode: startNode SourceObject [1]	The source object from which the path is followed to find the referenced data item.
PathMappingExecution [1] wasInformedBy PathExecution [1]	The source activity uses some output of the target activity.
PathExecution is specialisatie van OrchestrationActivity	An orchestration activity is an activity that is part of an IMX orchestration process.

§ 4.2.1.9 *SourceComplexDataItem*

Name	SourceComplexDataItem
Definition	A source complex data item is a complex data item that serves as input for an orchestration activity.
Is abstract	False

§ OVERVIEW RELATIONS

Relation name with multiplicity	Definition
SourceComplexDataItem [1] <u>consistsOf</u> SourceDataItem [2 .. *]	The constituent data item of the complex data item.
SourceComplexDataItem is specialisatie van <u>SourceDataItem</u>	A source data item is a data item that serves as input for an orchestration activity.

§ 4.2.1.10 Model

Name	Model
Definition	A model is the representation of a logical data model.
Date added	<date>
Is abstract	False

§ OVERVIEW ATTRIBUTES

Attribute name	Definition	Format	Card
<u>name</u>	The name of the model.	<u>Character-</u> <u>String</u>	1

§ OVERVIEW RELATIONS

Relation name with multiplicity	Definition
<u>Data Source</u> [1] <u>conformsTo</u> <u>Model</u> [1]	

§ 4.2.1.11 SourceSimpleDataItem

Name	SourceSimpleDataItem
Definition	A source simple data item is a simple data item that serves as input for an orchestration activity.
Is abstract	False

§ OVERVIEW ATTRIBUTES

Attribute name	Definition	Format	Card
<u>value</u>		<u>ValueType</u>	1

§ OVERVIEW RELATIONS

Relation name with multiplicity	Definition
SourceSimpleDataItem is specialisatie van SourceDataItem	A source data item is a data item that serves as input for an orchestration activity.

§ 4.2.1.12 PathMappingExecution

Name	PathMappingExecution
Definition	A path mapping execution is an orchestration activity that is part of the propertymapping execution.
Is abstract	False

§ OVERVIEW RELATIONS

Relation name with multiplicity	Definition
PathMappingExecution [1] <u>used</u> PathMapping [1]	The mapping element used in the orchestration activity.
PathMappingExecution [1] <u>wasInformedBy</u> PathExecution [1]	The source activity uses some output of the target activity.
PropertyMappingExecution [1] <u>wasInformedBy</u> PathMappingExecution [1 .. *]	The source activity uses some output of the target activity.
PathMappingExecution is specialisatie van Orchestration-Activity	An orchestration activity is an activity that is part of an IMX orchestration process.

§ 4.2.1.13 OrchestrationActivity

Name	OrchestrationActivity
Definition	An orchestration activity is an activity that is part of an IMX orchestration process.
Is abstract	False

§ 4.2.1.14 OrchestratedComplexDataItem

Name	OrchestratedComplexDataItem
Definition	An orchestrated complex data item is an orchestrated data item consisting of two or more other orchestrated data items.
Is abstract	False

§ OVERVIEW RELATIONS

Relation name with multiplicity	Definition
OrchestratedComplexDataItem [0 .. *] consistsOf OrchestratedDataItem [2 .. *]	The constituent data item of the complex data item.
OrchestratedComplexDataItem is specialisatie van OrchestratedDataItem	An orchestrated data item is a data item that is generated/obtained by orchestrating source data.

§ 4.2.1.15 PropertyMappingExecution

Name	PropertyMappingExecution
Definition	A Property mapping execution is an orchestration activity that pertains to the execution of a property mapping.
Date added	<date>
Is abstract	False

§ OVERVIEW RELATIONS

Relation name with multiplicity	Definition
PropertyMappingExecution [1] wasInformedBy Path-MappingExecution [1 .. *]	The source activity uses some output of the target activity.
PropertyMappingExecution [0 .. *] used: used Property-Mapping [1]	The mapping element used in the orchestration activity.
OrchestratedDataItem [0 .. *] wasGeneratedBy: was-GeneratedBy PropertyMappingExecution [1]	The activity which generated (or produced) the orchestrated data item.
PropertyMappingExecution is specialisatie van OrchestrationActivity	An orchestration activity is an activity that is part of an IMX orchestration process.

§ 4.2.1.16 OrchestratedSimpleDataItem

Name	OrchestratedSimpleDataItem
Definition	An orchestrated simple data item is an orchestrated data item that is non-divisible and contains a literal value.
Is abstract	False

§ OVERVIEW ATTRIBUTES

Attribute name	Definition	Format	Card
<u>value</u>	The value of the data item.	<u>ValueType</u>	1

§ OVERVIEW RELATIONS

Relation name with multiplicity	Definition
OrchestratedSimpleDataItem is specialisatie van <u>OrchestratedDataItem</u>	An orchestrated data item is a data item that is generated/obtained by orchestrating source data.

§ 4.2.2 Keuzen

§ 4.2.2.1 ValueType

Name	ValueType
-------------	-----------

§ OVERZICHT KEUZE ELEMENTEN

Keuze element	Definition	Format	Card
<u>boolean</u>		<u>Boolean</u>	1
<u>integer</u>		<u>Integer</u>	1
<u>lijn</u>		GM_Curve	1
<u>punt</u>		GM_Point	1
<u>real</u>		<u>Real</u>	1
<u>string</u>		<u>Character-</u> <u>String</u>	1
<u>vlak</u>		GM_Surface	1

§ 4.2.3 Attribuu- en relatiesoort details

§ 4.2.3.1 Objecttype details

§ 4.2.3.1.1 ORCHESTRATEDDATAITEM

Relatiesoort details [OrchestratedDataItem](#) wasGeneratedBy

Name	wasGeneratedBy
Definition	The activity which generated (or produced) the orchestrated data item.
Nillable	False
Multiplicity	1
Related objecttype	PropertyMappingExecution

Relatiesoort details [OrchestratedDataItem](#) subject

Name	subject
Definition	The subject of the data item.
Nillable	False
Multiplicity	1
Related objecttype	TargetObject

§ 4.2.3.1.2 SOURCEDATAITEM

Relatiesoort details [SourceDataItem](#) dataSource

Name	dataSource
Definition	The data source from which the source data item can be retrieved
Nillable	False
Multiplicity	1
Related objecttype	Data Source

Relatiesoort details [SourceDataItem](#) subject

Name	subject
Definition	The subject of a source data item.
Nillable	False
Multiplicity	1
Related objecttype	SourceObject

§ 4.2.3.1.3 SOURCERELATIONDATAITEM

Relatiesoort details [SourceRelationDataItem](#) value

Name	value
Definition	The value of the data item.
Nillable	False
Multiplicity	1
Related objecttype	SourceObject

§ 4.2.3.1.4 SOURCEOBJECT

Attribuutsoort details [SourceObject](#) type

Name	type
Definition	The type of the source object in the data source.
Nillable	False
Multiplicity	1
Format	CharacterString

Attribuutsoort details [SourceObject](#) identificatie

Name	identificatie
Definition	The identification of the source object in the data source.
Nillable	False
Multiplicity	1
Format	CharacterString

§ 4.2.3.1.5 DATA SOURCE

Attribuutsoort details [Data Source](#) name

Name	name
Definition	The name of the data source.
Nillable	False
Multiplicity	1
Format	CharacterString

Relatiesoort details [Data Source](#) conformsTo

Name	conformsTo
Nillable	False
Multiplicity	1
Related objecttype	Model

§ 4.2.3.1.6 ORCHESTRATEDRELATIONDATAITEM

Relatiesoort details [OrchestratedRelationDataItem](#) value

Name	value
Nillable	False
Multiplicity	1
Related objecttype	TargetObject

§ 4.2.3.1.7 TARGETOBJECT

Attribuutsoort details [TargetObject](#) identificatie

Name	identificatie
Definition	The identification of the target object
Nillable	False
Multiplicity	1
Format	CharacterString

Attribuutsoort details [TargetObject](#) Domain

Name	Domain
Nillable	False
Multiplicity	1
Format	CharacterString

§ 4.2.3.1.8 PATHEXECUTION

Relatiesoort details [PathExecution](#) references

Name	references
Definition	The source data item referenced by following the defined path starting from the start node.
Nillable	False
Multiplicity	0 .. *
Related objecttype	SourceDataItem

Relatiesoort details [PathExecution](#) used

Name	used
Definition	The mapping element used in the orchestration activity.
Toelichting	Een inhoudelijke toelichting op de definitie, ter verheldering of nadere duiding.
Nillable	False
Multiplicity	1
Related objecttype	Path

Relatiesoort details [PathExecution](#) startNode

Name	startNode
Definition	The source object from which the path is followed to find the referenced data item.
Nillable	False
Multiplicity	1
Related objecttype	SourceObject

§ 4.2.3.1.9 SOURCECOMPLEXDATAITEM

Relatiesoort details [SourceComplexDataItem](#) consistsOf

Name	consistsOf
Definition	The constituent data item of the complex data item.
Toelichting	A complex data item always consists out of two or more data items, which in turn can consists out of other data items.
Nillable	False
Multiplicity	2 .. *
Related objecttype	SourceDataItem

§ 4.2.3.1.10 MODEL

Attribuutsoort details [Model](#) name

Name	name
Definition	The name of the model.
Toelichting	Een inhoudelijke toelichting op de definitie, ter verheldering of nadere duiding.
Nillable	False
Multiplicity	1
Patroon	De verzameling van waarden die gegevens van deze attribuutsoort kunnen hebben, oftewel het waardenbereik, uitgedrukt in een specifieke structuur.
Format	CharacterString

§ 4.2.3.1.11 SOURCESIMPLEDATAITEM

Attribuutsoort details [SourceSimpleDataItem](#) value

Name	value
Nillable	False
Multiplicity	1
Format	ValueType

§ 4.2.3.1.12 PATHMAPPINGEXECUTION

Relatiesoort details [PathMappingExecution](#) used

Name	used
Definition	The mapping element used in the orchestration activity.
Nillable	False
Multiplicity	1
Related objecttype	PathMapping

Relatiesoort details [PathMappingExecution](#) wasInformedBy

Name	wasInformedBy
Definition	The source activity uses some output of the target activity.
Nillable	False
Multiplicity	1
Related objecttype	PathExecution

§ 4.2.3.1.13 ORCHESTRATEDCOMPLEXDATAITEM

Relatiesoort details [OrchestratedComplexDataItem](#) consistsOf

Name	consistsOf
Definition	The constituent data item of the complex data item.
Toelichting	A complex data item always consists out of two or more data items, which in turn can consists out of other data items.
Nillable	False
Multiplicity	2 .. *
Related objecttype	OrchestratedDataItem

§ 4.2.3.1.14 PROPERTYMAPPINGEXECUTION

Relatiesoort details [PropertyMappingExecution](#) wasInformedBy

Name	wasInformedBy
Definition	The source activity uses some output of the target activity.
Toelichting	Een inhoudelijke toelichting op de definitie, ter verheldering of nadere duiding.
Nillable	False
Multiplicity	1 .. *
Related objecttype	PathMappingExecution

Relatiesoort details [PropertyMappingExecution](#) used

Name	used
------	------

Definition	The mapping element used in the orchestration activity.
Nilable	False
Multiplicity	1
Related objecttype	PropertyMapping

§ 4.2.3.1.15 ORCHESTRATEDSIMPLEDATAITEM

Attribuutsoort details [OrchestratedSimpleDataItem](#) value

Name	value
Definition	The value of the data item.
Nilable	False
Multiplicity	1
Format	ValueType

§ 4.2.3.2 Keuze

§ 4.2.3.2.1 KEUZE VALUETYPE

Keuze element [ValueType](#) boolean

Name	boolean
Multiplicity	1
Format	Boolean

Keuze element [ValueType](#) integer

Name	integer
Multiplicity	1
Format	Integer

Keuze element [ValueType](#) lijn

Name	lijn
Multiplicity	1
Format	GM_Curve

Keuze element [ValueType](#) punt

Name	punt
Multiplicity	1
Format	GM_Point

Keuze element [ValueType](#) real

Name	real
Multiplicity	1

Format	<u>Real</u>
--------	-------------

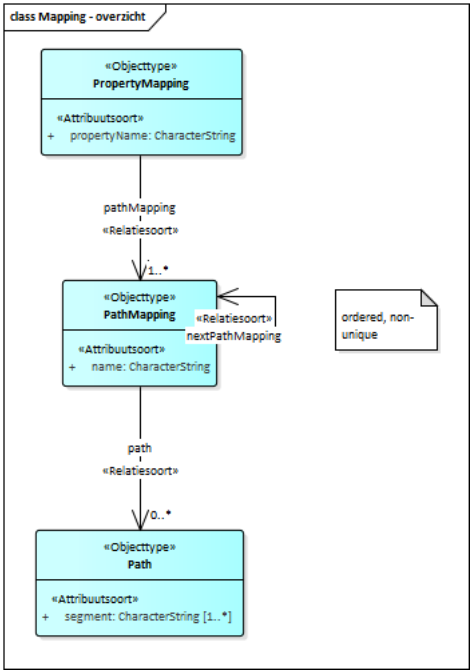
Keuze element ValueType string

Name	string
Multiplicity	1
Format	<u>CharacterString</u>

Keuze element ValueType vlak

Name	vlak
Multiplicity	1
Format	GM_Surface

§ 4.3 View IMX Mapping model



Mapping

§ 4.3.1 Objecttypes

§ 4.3.1.1 PathMapping

Name	PathMapping
Definition	A path mapping is a set of declarations which express references to data items of data objects of source data sources and actions on those.
Is abstract	False

§ OVERVIEW ATTRIBUTES

Attribute name	Definition	Format	Card
<u>name</u>	The name of the path mapping.	<u>Character-</u> <u>String</u>	1

§ OVERVIEW RELATIONS

Relation name with multiplicity	Definition
PathMapping [1] <u>path: path</u> Path [0 .. *]	
PathMapping [1] <u>nextPathMapping</u> PathMapping [1]	The subsequent path mapping to be executed as part of a property mapping.
PathMappingExecution [1] <u>used</u> PathMapping [1]	The mapping element used in the orchestration activity.
PropertyMapping [1] <u>pathMapping: pathMapping</u> Path-Mapping [1 .. *]	

§ 4.3.1.2 Path

Name	Path
Definition	A path, or path expression, is a set of declarations which, conceptually, represent a possible route through a graph of data items. A path is comprised of an ordered, non-unique sequence of segments. Each segment represents a step in that route.
Is abstract	False

§ OVERVIEW ATTRIBUTES

Attribute name	Definition	Format	Card
<u>segment</u>	A segment is a string that represents a property reference.	<u>Character-</u> <u>String</u>	1 .. *

§ OVERVIEW RELATIONS

Relation name with multiplicity	Definition
PathExecution [1] <u>used: used</u> Path [1]	The mapping element used in the orchestration activity.
PathMapping [1] <u>path: path</u> Path [0 .. *]	

§ 4.3.1.3 PropertyMapping

Name	PropertyMapping
Definition	A property mapping is a set of declarations which express the mapping of a target object type property within an object type mapping.
Is abstract	False

§ OVERVIEW ATTRIBUTES

Attribute name	Definition	Format	Card
propertyName	The name of the property in the target model which is the target of this property mapping.	Character- String	1

§ OVERVIEW RELATIONS

Relation name with multiplicity	Definition
PropertyMapping [1] pathMapping : pathMapping Path-Mapping [1 .. *]	
PropertyMappingExecution [0 .. *] used : used Property-Mapping [1]	The mapping element used in the orchestration activity.

§ 4.3.2 Attribuut- en relatiesoort details

§ 4.3.2.1 Objecttype details

§ 4.3.2.1.1 PATHMAPPING

Attribuutsoort details [PathMapping](#) name

Name	name
Definition	The name of the path mapping.
Toelichting	Een inhoudelijke toelichting op de definitie, ter verheldering of nadere duiding.
Nilable	False
Multiplicity	1
Patroon	De verzameling van waarden die gegevens van deze attribuutsoort kunnen hebben, oftewel het waardenbereik, uitgedrukt in een specifieke structuur.
Format	CharacterString

Relatiesoort details [PathMapping](#) path

Name	path
Nillable	False
Multiplicity	0 .. *
Related objecttype	Path

Relatiesoort details [PathMapping](#) nextPathMapping

Name	nextPathMapping
Definition	The subsequent path mapping to be executed as part of a property mapping.
Nillable	False
Multiplicity	1
Related objecttype	PathMapping

§ 4.3.2.1.2 PATH

Attribuutsoort details [Path](#) segment

Name	segment
Definition	A segment is a string that represents a property reference.
Nillable	False
Multiplicity	1 .. *
Format	CharacterString

§ 4.3.2.1.3 PROPERTYMAPPING

Attribuutsoort details [PropertyMapping](#) propertyName

Name	propertyName
Definition	The name of the property in the target model which is the target of this property mapping.
Toelichting	Een inhoudelijke toelichting op de definitie, ter verheldering of nadere duiding.
Nillable	False
Multiplicity	1
Patroon	De verzameling van waarden die gegevens van deze attribuutsoort kunnen hebben, oftewel het waardenbereik, uitgedrukt in een specifieke structuur.
Format	CharacterString

Relatiesoort details [PropertyMapping](#) pathMapping

Name	pathMapping
Nillable	False
Multiplicity	1 .. *
Related objecttype	PathMapping

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§ A.2 Terms defined by reference

§ B. References

§ B.1 Informative references

[IMX-Metadata]

[*IMX - Metadata basic principles*](#). Geonovum. W3C Editor's Draft. URL: <https://geonovum.github.io/IMX-Metadata>

[IMX-ModelMapping]

[*IMX Model Mapping*](#). Geonovum. W3C Editor's Draft. URL: <https://geonovum.github.io/IMX-ModelMapping>

[Property-Stereotype-for-Metadata]

[*Property Stereotype for Metadata*](#). Publisher-1. V1.0. URL: https://shapechange.net/wp-content/uploads/2019/12/UGAS19-D100_property_stereotypes.pdf

[prov-o]

[*PROV-O: The PROV Ontology*](#). Timothy Lebo; Satya Sahoo; Deborah McGuinness. W3C. 30 April 2013. W3C Recommendation. URL: <https://www.w3.org/TR/prov-o/>